

BUSINESS ANALYTICS & CONSULTING CASE STUDY

Grocery Gateway

Revenue Mix, Profitability, and Working-Capital Risk at an Independent Neighbourhood Grocery Store

Case study prepared by Aanchal Routela
Business Analyst | Aspiring Product Manager

About this document

This case study is adapted from an independent Business Data Management (BDM) capstone project conducted on a real, small-format retail store. It has been restructured into a business-analytics portfolio format for review by product, analytics, and consulting audiences. All figures, charts, and findings are drawn directly from the original data collection and analysis; no metrics, outcomes, or business impact have been added or fabricated. Where the original analysis drew conclusions not fully supported by the underlying data, this version reframes them explicitly as hypotheses.

Executive Summary

Vinod Store is a small, independently owned grocery store that has served a residential neighbourhood in South Delhi since 2003. It operates the way most Indian neighbourhood “kirana” stores do: paper-based billing, informal credit for regular customers, and no formal analytics. This project set out to build a lightweight analytics layer on top of the store's existing records — sales, inventory, and credit — to answer three practical questions the owner could not easily answer himself: is the store growing, where is money being lost, and what should be prioritised first.

Working directly with the store owner, I manually collected and structured five months of daily sales, inventory, and credit data (January–May 2024), together with the store's available annual sales totals from 2018–22. I organised this into a working Excel model, computed category-level revenue and margins, and built a set of visualisations to surface patterns in sales, category performance, uncollected credit, and perishable-item wastage.

Headline observations

- Average daily sales rose from ₹20,564 in January to ₹26,410 in May 2024 — a ~28% increase over the five-month window, after a dip in February.
- Perishable Items alone accounted for ₹77,425 of charted category revenue — roughly half of the total — with Grocery (₹28,259) and Beverages (₹24,737) next; Hygiene Products and Extra Items together contributed under 5%.
- Both average sales and outstanding (unpaid) credit rose steadily from January to May 2024 in absolute terms.
- Wastage of perishable items reduced profit in every period charted, and recorded loss-days were concentrated — 7 of 10 total loss-days fell within a single month.

What this document contains

The sections that follow walk through the business context, the data and method, the analysis and insights (each separating what was observed from what is interpretation), the resulting recommendations, a simple impact-vs-effort prioritisation, expected — not measured — impact, and the limitations of a single-store, five-month, manually-collected dataset. A closing section summarises my personal contribution and what I took away from the project.

Scope note

The recommendations in this document were formulated as part of the original analysis but were **not implemented** during the study period. No post-implementation results, A/B tests, or measured business impact exist. Any impact discussed below is a directional hypothesis, not a measured outcome.

1. Business Context

Vinod Store is a compact, independently run grocery and convenience store that has served its local community in South Delhi since 2003. Like most small-format Indian retail (“kirana”) stores, it competes on proximity, relationships, and informal credit rather than scale, and it runs without any formal point-of-sale or analytics system.

I had direct, on-site access to the store because of its proximity to my residence, which allowed me to collect granular sales, inventory, and credit data in person over an extended period, in close collaboration with the store owner, Mr. Vinod. The objective was to treat Vinod Store as a representative case of a small independent grocery business, and to use its own operational data to surface patterns the owner could act on.

This is a business analytics and consulting engagement, not a technology or automation project: the deliverable is a structured analysis of existing store data and a set of prioritised, practical recommendations — not a new tool, app, or system.

2. Business Problem

Three recurring, owner-reported challenges framed the scope of the analysis:

- **Customer acquisition and sales retention** — no clear, data-backed view of whether footfall and sales were growing, shrinking, or seasonal.
- **Uncollected debt** — a meaningful share of sales is extended as informal credit to regular customers, with no systematic tracking of how much is outstanding or how it moves month to month.
- **Inventory obsolescence and wastage** — perishable items (dairy and similar short-shelf-life goods) are a core part of the business, but spoilage and unsold stock erode their profitability in ways the owner could not quantify.

The analysis was designed to quantify each of these three problems using the store's own records, rather than assumptions, and to translate the findings into concrete, prioritised actions.

3. Data & Methodology

3.1 Data collection

Sales, inventory, and credit data were collected manually from the store's paper records through regular on-site visits between January and May 2024, and organised into a structured Excel workbook (separate tabs for sales, debt, perishables, analysis, and charts). Entries were cross-checked with Mr. Vinod to catch missing or erroneous records. In addition, the store's available annual sales totals for 2018–2022 were used to provide longer-run context.

3.2 What was tracked

- Daily and monthly sales revenue, split across seven product categories: Perishable Items, Grocery, Beverages, Eatables, Oils, Hygiene Products, and Extra Items.
- Cost of goods sold, used to derive gross profit by category and by month.
- Fixed monthly costs — recorded as Transport (₹4,000), Rent (₹2,550), and Electricity (₹500) — distributed evenly across the days of the month and deducted from gross profit to arrive at net profit ($\text{Net Profit} = \text{Gross Profit} - \text{Daily Fixed Cost share}$).
- Credit extended, outstanding (unpaid) debt, and debt returned, tracked monthly.
- Perishable-item purchase, sales, and wastage quantities, and the number of days per month on which a wastage-related loss was recorded.

3.3 Method

The analysis combined descriptive statistics (averages, totals, category shares) with comparative time-series analysis across months and years, and category-level breakdowns of revenue and margin. Findings were visualised using bar, line, pie, and tree-map charts built directly from the Excel model, which are reproduced in Section 4.

A note on rigour

This is an observational, single-store dataset built from manually recorded transactions over a five-month primary window (plus five non-contiguous annual totals). It supports *descriptive* and *associative* findings, not causal ones. Every claim below is labelled as either an observed finding (directly read from the data/chart) or an interpretation (a plausible explanation that has not been independently tested against the data).

4. Analysis & Insights

Each insight below separates the observed finding (read directly from the chart/data) from the interpretation (a plausible explanation). Where the original analysis drew a conclusion the underlying chart does not clearly support, that is flagged explicitly.

4.1. Sales trend, January–May 2024

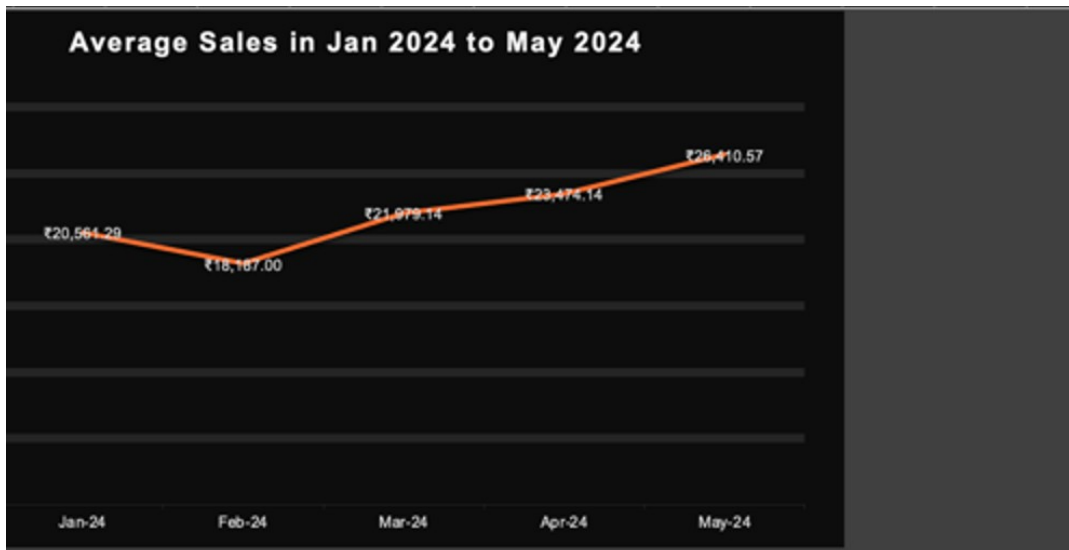


Fig. 1 — Average daily sales, Jan–May 2024 (source: store sales records).

Observed finding: Average daily sales moved from ₹20,564.29 (Jan) to ₹18,187.00 (Feb), then rose each month to ₹21,979.14 (Mar), ₹23,474.14 (Apr), and ₹26,410.57 (May) — a net increase of about 28% from January to May, after an initial dip.

Interpretation: The February dip followed by a sustained three-month recovery is consistent with a seasonal pattern rather than a declining business. The original report attributes the March uptick to the Holi festival and the April–May rise to summer demand for cold beverages and snacks; this is a plausible hypothesis, but the charted data is a single store-wide sales total, not broken down by category or week, so the festival/seasonal driver cannot be directly verified from this dataset.

4.2. Long-run annual sales, 2018–2022

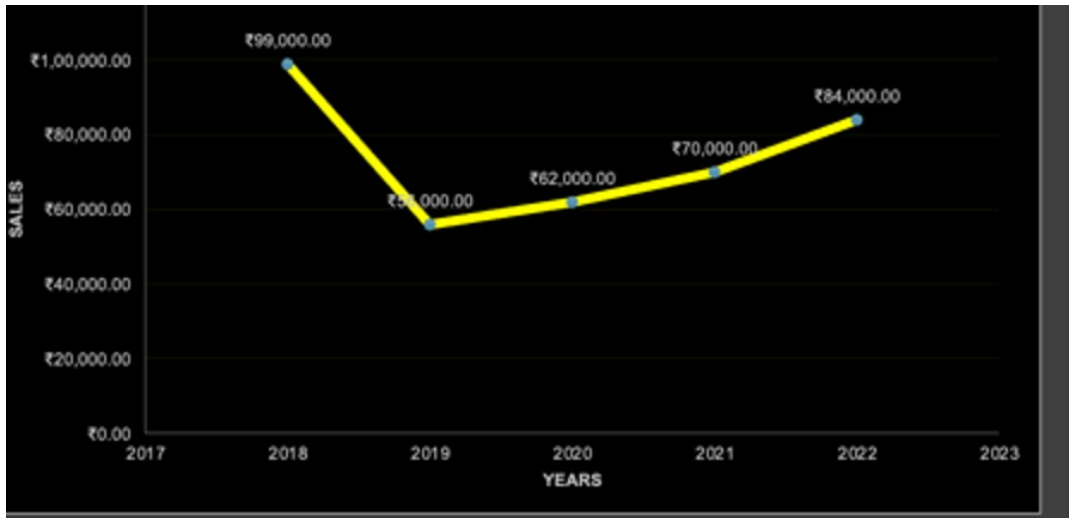


Fig. 2 — Annual sales, 2018–2022 (source: store historical records).

Observed finding: Annual sales fell sharply from ₹99,000 (2018) to roughly ₹58,000 (2019) — a drop of about 41% — then recovered gradually each year: ₹62,000 (2020), ₹70,000 (2021), and ₹84,000 (2022). By 2022, sales had not yet returned to the 2018 level.

Interpretation: The original report offers two different explanations for the 2019 dip in different sections — “covid” in one place and “increased competition or supply chain disruption” in another — which are inconsistent (COVID-related disruption in India was concentrated from 2020 onward, not 2019). The cause of the 2019 dip should therefore be treated as unverified. What the chart does support is a clear multi-year recovery trend through 2022, and a gap between the 2024 monthly detail and this annual series (no 2023 figure is available), which limits direct comparison between the two.

4.3. Revenue mix by product category

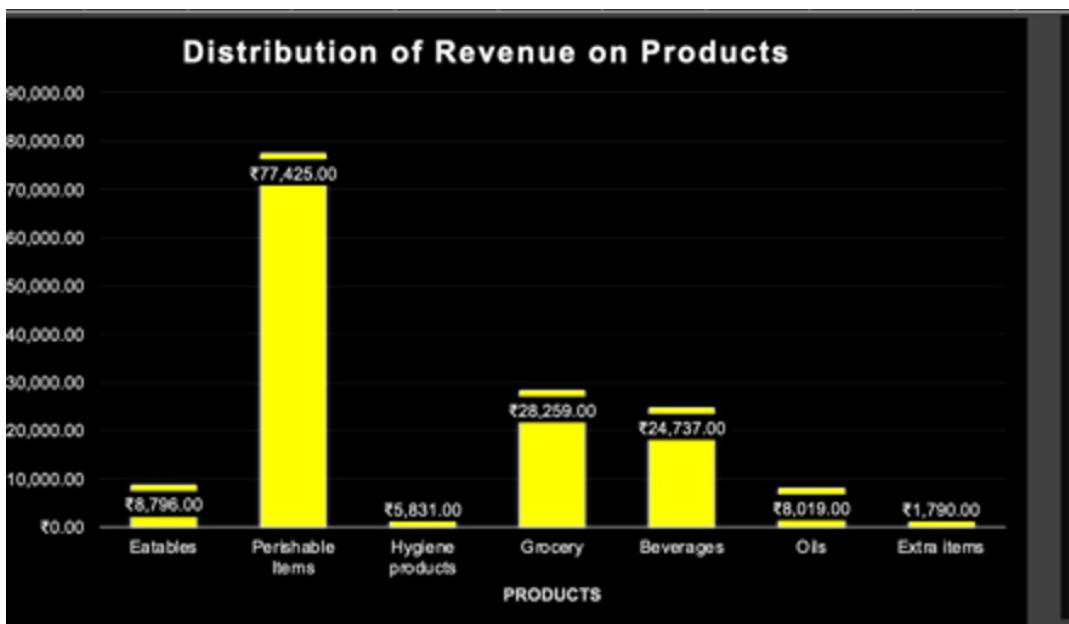


Fig. 3 — Revenue by product category (source: store sales records).

Observed finding: Across the charted period, category revenue was: Perishable Items ₹77,425, Grocery ₹28,259, Beverages ₹24,737, Eatables ₹8,796, Oils ₹8,019, Hygiene Products ₹5,831, and Extra Items ₹1,790 — a charted total of ₹154,857. Perishable Items alone represent roughly half of that total.

Interpretation: Revenue is heavily concentrated in two essential, high-turnover categories (Perishable Items and Grocery), which together make up about 68% of charted revenue. Hygiene Products and Extra Items combined contribute under 5%. This is consistent with the store's role as a source of daily essentials rather than a destination for discretionary categories.

4.4. Category-wise profit margin

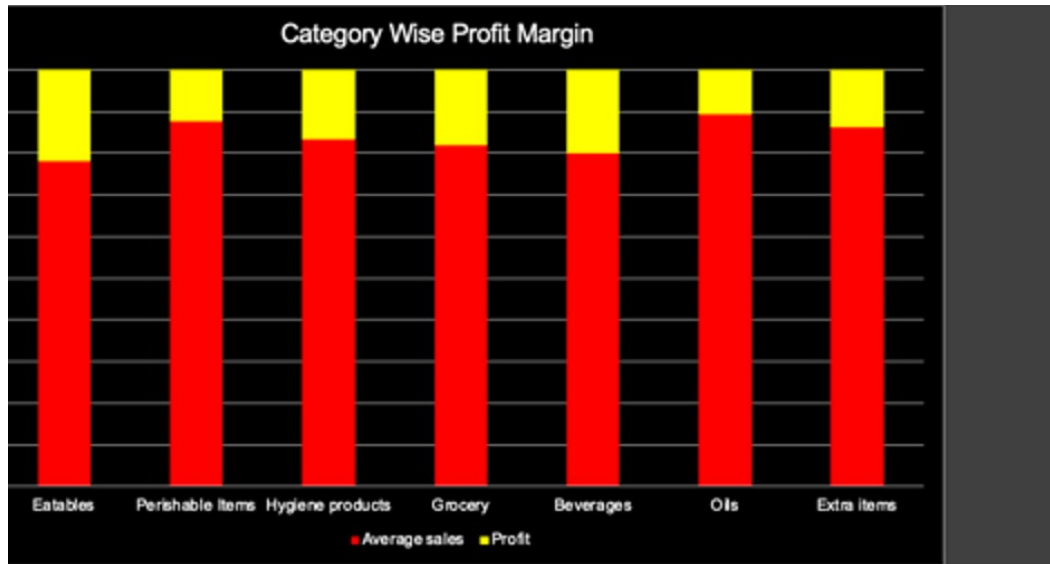


Fig. 4 — Category-wise average sales vs. profit share (source: store sales/COGS records).

Observed finding: The chart plots average sales against profit for each category as a share of the bar. Visually, Eatables shows the largest profit share of the seven categories; Oils, Perishable Items, and Extra Items show comparatively smaller profit shares. Exact percentage values are not numerically labelled on the source chart.

Interpretation: This does not clearly support the original report's narrative claim that "Oils and Beverages carry the highest margins." Visually, Oils shows one of the smaller profit shares in this chart, not one of the largest. Because the chart has no numeric labels, this should be treated as a hypothesis to be re-tested with exact category-level margin figures, rather than a confirmed finding — the claim is not removed, but it is downgraded from "result" to "unverified."

4.5. Uncollected (credit) debt

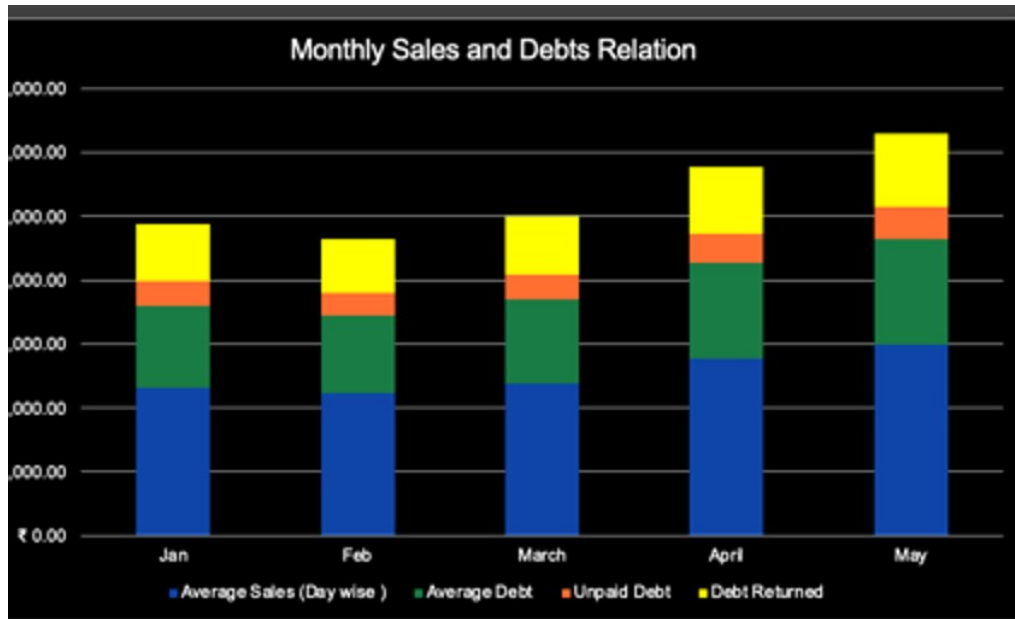


Fig. 5 — Monthly sales and debt relationship, Jan–May 2024 (source: store credit records).

Observed finding: The stacked chart shows four components — average sales, average debt, unpaid debt, and debt returned — all rising in absolute terms from January to May 2024.

Interpretation: Rising unpaid debt alongside rising sales is not, by itself, evidence of “controlled” debt — the original report's conclusion that debt is “effectively managed” is not directly supported by this chart, since the unpaid-debt segment visually grows rather than shrinks, and the chart does not show debt as a share of sales. Separately, the store's own records indicate that a small group of regular, credit-purchasing customers accounts for a large share of the outstanding balance; no customer-level chart was available to independently verify this concentration, so it is reported here as a finding from the underlying records rather than something visualised.

4.6. Perishable-item wastage and profit

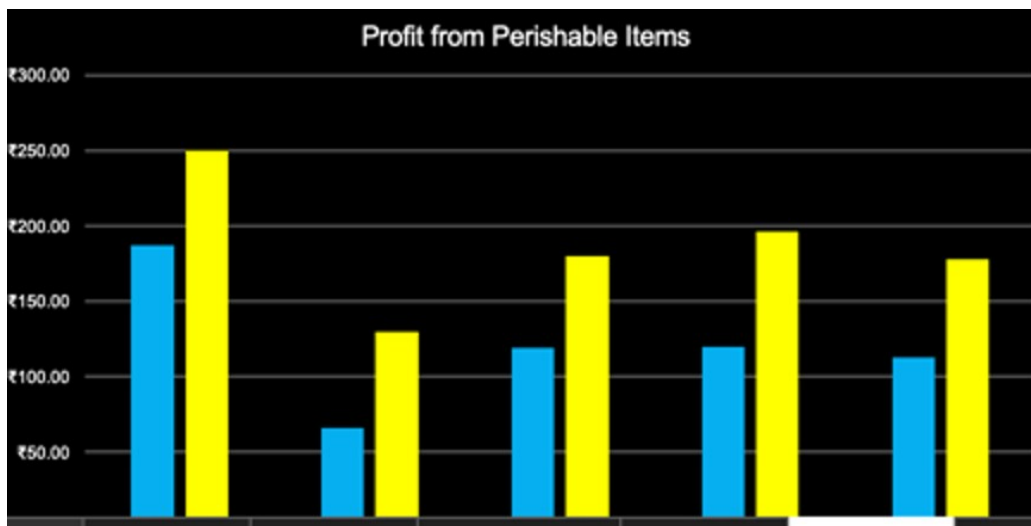


Fig. 6 — Average profit from perishable items, with vs. without wastage (source: store perishables ledger).

Observed finding: Profit excluding wastage was higher than profit including wastage in every period charted, with the gap ranging roughly ₹15–₹65 depending on the period. Separately, a pie chart of loss-days per month shows values of 0, 1, 0, 2, and 7 across the five months — a total of 10 loss-days, with one single month (identified in the source text as February) accounting for 7 of those 10.

Interpretation: Wastage is a recurring drag on perishable-item profit rather than a one-off event — it appears in every period, not just one. However, loss-days are heavily concentrated in a single month rather than spread evenly, which points to a specific operational or forecasting gap in that period rather than a stable weekly pattern. The original report's separate claim that wastage “primarily occurs midweek (Wednesday/Thursday)” cannot be verified from this chart, since the pie chart breaks losses down by month, not by day of week — that specific claim is therefore not carried forward as a finding.



Fig. 7 — Number of days per month with a recorded perishable-item loss (source: store perishables ledger).

5. Business Implications

Taken together, the six insights point to three practical implications for the business:

- **Profitability is structurally exposed.** Revenue is concentrated in two essential, high-turnover, comparatively lower-margin categories (Perishable Items and Grocery), so overall profitability is sensitive to cost control and to whether higher-margin categories can be grown — though which categories are actually higher-margin needs re-verification (see 4.4).
- **Working capital is tied up in credit sales.** Unpaid debt is rising in absolute terms alongside sales; without a clearer view of debt as a share of sales, the store cannot easily tell whether this is a manageable, growth-proportional pattern or an emerging cash-flow risk.
- **Wastage is a repeatable, not incidental, profit leak.** Every period charted shows a wastage-related profit gap, and losses cluster in specific months — suggesting a fixable forecasting or ordering issue rather than random spoilage.

6. Recommendations

The table below consolidates the recommendations from the original analysis into five themes, each tied explicitly to the finding and interpretation that motivates it. All recommendations are proposals; none were implemented or tested within the scope of this project.

Finding	Interpretation	Recommendation
Sales dip in Feb 2024, then steady recovery through May; multi-year recovery trend 2019–2022.	<i>Sales pattern may be seasonal.</i>	Plan seasonal/festival promotions and midweek deals around historically slower periods; use simple month-over-month tracking (in place of forecasting software) to anticipate demand swings.
Perishables + Grocery = ~68% of revenue; Hygiene/Extra Items under 5%.	<i>Underperforming categories are getting limited visibility or promotion.</i>	Pilot small bundle/cross-promotion offers pairing a slow-moving category (e.g., Hygiene) with a high-traffic staple, and track category revenue before/after.
Category profit-share ranking in the chart does not match the original report's Oils/Beverages claim.	<i>The true higher-margin categories are unclear without exact figures.</i>	Before promoting any category as "higher-margin," recompute exact per-category margin percentages from the underlying sales/COGS data, then target the promotion at whichever categories that recomputation actually supports.
Unpaid debt rises in absolute terms alongside sales, Jan–May 2024; concentrated among repeat credit customers per store	<i>Credit exposure may be growing rather than staying flat.</i>	Introduce simple, low-friction credit controls: a per-customer credit ceiling, an early-payment discount, and regular (e.g.,

Finding	Interpretation	Recommendation
records.		weekly) reminders for balances outstanding beyond an agreed period.
Wastage reduces perishable profit in every period charted; loss-days concentrated in one month (7 of 10).	<i>Losses look operational (ordering/forecasting) rather than random.</i>	Introduce basic stock-rotation practice (first-in, first-out) and markdown/flash-sale pricing for perishables nearing expiry, and review ordering quantities specifically for the month(s) that showed concentrated losses.

7. Prioritisation

The five recommendation themes are placed below on a simple impact-vs-effort view. This prioritisation is my own synthesis, added for this case study to reflect how I would sequence the work as a product/business analyst — it was not part of the original academic report.

Recommendation theme	Impact	Effort	Sequencing rationale
Credit control (limits, reminders, early-payment discount)	High	Low	<i>Do first — addresses working-capital risk with mostly policy changes, no new tooling.</i>
Perishable stock rotation & markdown near expiry	High	Low-Medium	<i>Do first/second — directly targets a recurring, quantified profit leak.</i>
Re-verify category margins before acting on them	Medium	Low	<i>Do early — cheap to do, and de-risks the margin-promotion recommendation below.</i>
Seasonal/midweek promotions & simple demand tracking	Medium	Medium	<i>Sequence after the above — useful, but depends on having a season's worth of tracked data to act on.</i>
Cross-promotion / bundling for under-performing categories	Low-Medium	Medium	<i>Test small — lowest evidence base of the five; pilot before scaling.</i>

8. Expected Impact

Important

None of the recommendations above were implemented within the scope of this project. The statements below are **hypothesised, directional expectations**, not measured results. No before/after experiment, pilot, or business-impact data exists for this store.

- **Credit controls:** directionally expected to slow the growth of unpaid debt relative to sales, improving cash available for restocking — not quantified, since no baseline debt-to-sales ratio was computed from the current chart.
- **Perishable stock rotation & markdowns:** directionally expected to narrow the with-wastage/without-wastage profit gap seen in Fig. 6, and to reduce the concentration of loss-days seen in Fig. 7 — not quantified without a follow-up measurement period.
- **Margin re-verification:** expected to prevent a mis-targeted promotion (i.e., promoting a category assumed to be high-margin that may not be) — a risk-avoidance benefit rather than a revenue uplift.
- **Seasonal/midweek promotions:** directionally expected to smooth the Jan–May sales pattern seen in Fig. 1, particularly the February dip — not quantified.
- **Category bundling/cross-promotion:** directionally expected to lift revenue share for Hygiene Products and Extra Items from its current sub-5% share — lowest-confidence expectation of the five, and the first candidate for a small test before wider rollout.

9. Limitations

- Single-store case study: findings reflect one independent grocery store in one neighbourhood and may not generalise to other stores or formats.
- Short primary window: the detailed, daily-level analysis covers five months (Jan–May 2024); the annual series (2018–22) is five non-contiguous yearly totals with no 2023/2024 annual figure to bridge the two.
- Manually collected data: sales, inventory, and credit records were captured by hand from paper records, which carries a risk of entry error or omission that could not be independently audited for this case study.
- No controlled comparison: all findings are observational and associative; none of the seasonal, festival, or weather-driven explanations offered in the original analysis were tested against category- or week-level data, because that level of breakdown was not part of the charted dataset.
- Chart readability: several source charts (e.g., category profit margin, monthly sales-debt relation) do not carry numeric axis labels for every series, which limits how precisely their patterns can be quoted — this is flagged inline wherever it applies.
- No customer-level data: the claim that a small group of customers holds most of the outstanding debt is drawn from the underlying store records described in the original report, not from a customer-level chart available for this case study.
- Recommendations are untested: all recommendations and impact statements in Sections 6–8 are proposals and hypotheses; no pilot, A/B test, or post-implementation data exists.

10. My Contribution & Learnings

10.1 What I did

- Collected sales, inventory, and credit data on-site over five months through regular visits and direct collaboration with the store owner.
- Designed and built the Excel data model: category-level revenue, cost of goods sold, fixed costs, net profit, and credit tracking, organised across linked sheets.
- Built the bar, line, pie, and tree-map visualisations used to analyse sales trends, category performance, debt, and perishable wastage.
- Performed the descriptive and comparative analysis, and authored the original findings and recommendations.
- For this case study, re-examined each original finding against its underlying chart, separated observed findings from interpretation, reframed unsupported causal claims as hypotheses, and added the prioritisation framework in Section 7.

10.2 What I learned

- Real-world small-business data is messy by default — a meaningful share of the effort was in structuring and cleaning manually recorded transactions before any analysis could begin.
- A finding and an explanation for that finding are not the same thing — revisiting this project specifically to separate “what the chart shows” from “why I think it shows that” surfaced at least one claim (category margin ranking) that the chart did not actually support.
- Recommendations for a non-technical, resource-constrained stakeholder need to be sequenced, not just listed — which is why this version adds an explicit impact-vs-effort prioritisation that the original report did not include.
- Precise, labelled charts matter for reproducibility — several source charts in this project lacked full numeric labelling, which limited how confidently some patterns could be restated later.